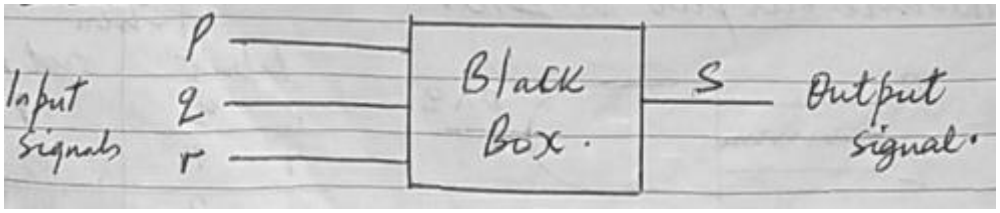
LOGIC CIRCUITS

NOTES COMPILED BY MUSYOKA D. M

- Propositional logic can be applied in design of computer hardware and thus in this subtopic, we give a brief introduction to this.
- A logic circuit/digital circuit receives input signals $p_1, p_2, \ldots, p_n$ each a bit (either 0- OFF or 1 – ON) and produces output signals $s_1, s_2, \ldots, s_n$, each a bit.
- In this section, we look at Logic circuits with a single output signal.
- The basic electronic components of a digital system are called digital logic circuits.
- "LOGIC" – indicates the important role of logic in design of such circuits.
- "DIGITAL" – Indicates that the circuits process discrete or separate signals as opposed to continuous ones.
- Electrical engineers use language of logic i.e., TRUE and FALSE to refer to values of signals produced by an electronic switch but these symbols have been replaced by symbols 0 and 1 – called bits.

Black boxes and gates

- Combinations of signal bits (1's and 0's) can be transformed to other combinations of signal bits (1's and 0's) using various circuits.
- Computer engineers think of such basic circuits as black boxes
- The box contains – a detailed implementation of the circuit.



- Here we have 3 input signals. Each signal can take 1 or 0, so there are 8 possible combinations of input signals.

p	q	r	s
1	1	1	1
1	1	0	0
1	0	1	0
1	0	0	1
0	1	1	0
0	1	0	1
0	0	1	1
0	0	0	0

This is a possible correspondence.

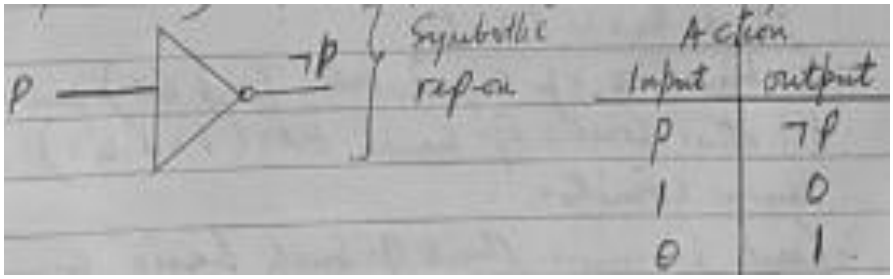
BASIC LOGIC GATES

An efficient way to design more complicated circuits is to build them by connecting less complicated black box circuits.

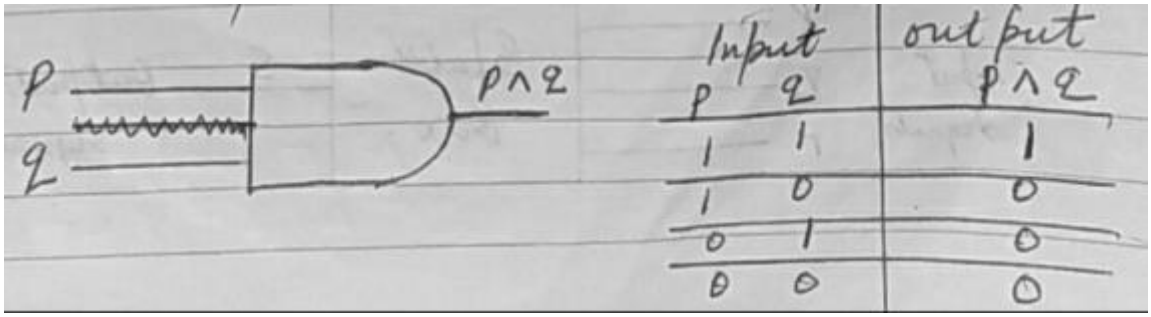
Three such circuits are: -

1. **NOT**

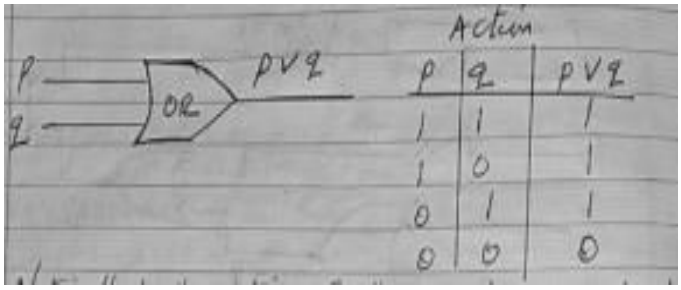
A NOT – gate /Inverter – is a circuit with one input signal e.g., p and one output signal e.g. $(\neg p)$ i.e., if the input is 1, output is 0 and conversely if input is 0, the output is 1.



2. **AND – gate** – has 2 input signals and one output signal. If both input signals are 1, the output is 1, otherwise the output is zero.



3. **OR – gate** – has two input signals and one output signal. If both input signals are 0, then the output is zero, otherwise the output is 1.



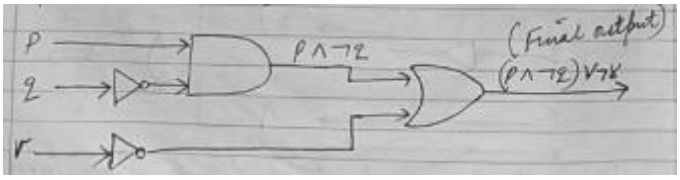
Notice that the actions of the 3 gates correspond exactly to those of logical connectives $\neg$, $\wedge$ *and* $\vee$ on statements, If 1 is identified with T and 0 is identified with F.

Combinations of these 3 basic gates can be used to build more complicated circuits called – **combinational/combinatorial circuits.**

Rules for combinational circuits

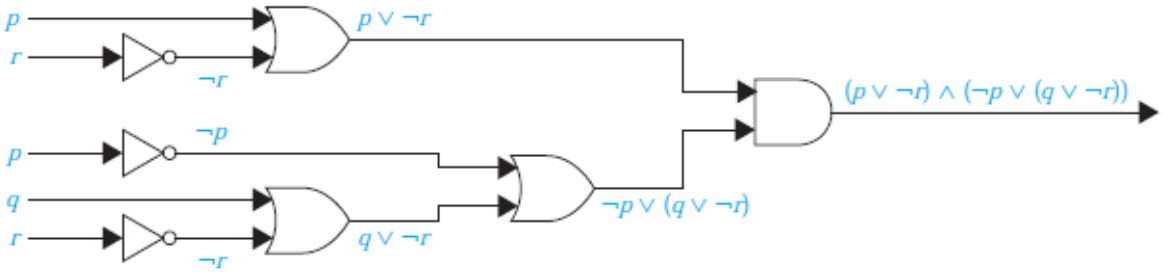
1. Never combine two input wires
2. A single input wire can be split part way and used as input for 2 separate gates.
3. An output wire can be used as input to another gate.
4. No output of a gate can be eventually be fed back to the same gate.

Example



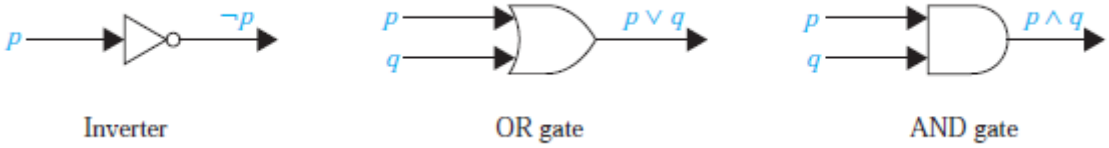
Example

Build a digital circuit that produces the output $(p \vee\sim r) \wedge (\sim p \vee (q \vee\sim r))$ when given the input bits p, q *and* r.



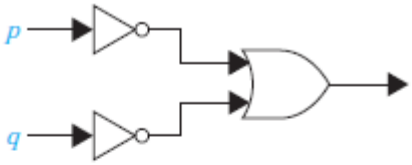
Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $(p \wedge \neg r) \vee (\neg q \wedge r)$ from input bits p, q, and r.

Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces the output $((\neg p \vee \neg r) \wedge \neg q) \vee (\neg p \wedge (q \vee r))$ from input bits p, q, and r.

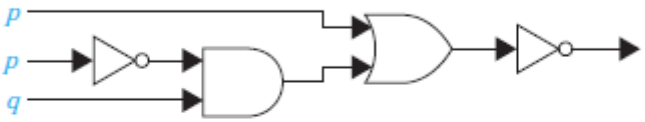


Find the output of each of these combinatorial circuits.

a)

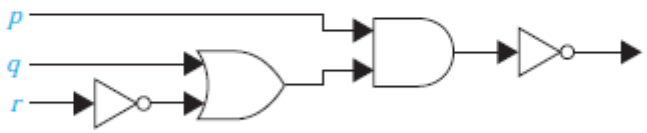


b)



Find the output of each of these combinatorial circuits.

a)



b)

